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LEENA NANDAN, *Secretary, Union Ministry of Environment, Forest & Climate Change*

of hand and calling for sustained action, the Narendra Modi government launched the National Clean Air Programme in January 2019 to improve the air quality in 131 cities where the air pollution levels exceeded permissible levels and bring them down to the National Ambient Air Quality Standard (NAAQS). Close to Rs 20,000 crore until FY26 was allotted to the project.

The Centre, along with independent experts from institutions like the IITs, worked with state governments and civic bodies in these cities to come up with individual air-shed plans to monitor, analyse and control the main sources of pollution, whether vehicular emissions, construction and road dust, garbage disposal or industrial pollution.

Now midway into the plan, the Centre claims that 90 cities have shown improved air quality, and 15 of them have met the required air quality standards. "The whole approach to control air pollution in urban centres has gathered momentum only in the past few years," says Leena Nandan, secretary, Union ministry of environment, forest and climate change (MoEFCC). "We realised that we couldn't have a one-size-fits-all approach, so each city was asked to come up with an air-shed approach that has a technical appreciation of its unique circumstances and problems and how to deal with them. We've achieved much, but there is still a lot of distance to cover."

Meanwhile, an October 2022 report by McKinsey broke down the various factors contributing to pollution and deduced that the power sector accounted for 34 per cent, industry, including the construction sector, 28 per cent, agriculture 17.8 per cent and the transport segment 9 per cent of the problem of pollution. These may be ballpark estimates, and may well vary in individual air-shed regions. For instance, an ongoing study by IIT Kanpur, IIT Delhi and TERI shows that, at 30 per cent, the transport sector is one of the highest average percentage contributors to the PM2.5 levels in Delhi. But while the four polluting sectors—automobiles, agriculture, power and construction—may afflict individual cities and zones in varying degree, they remain the chief villains in India's air pollution woes. Combating them requires both short and long term measures. Here are a few of them.

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For a country as large as India, the sheer number of motor vehicles on its roads can be mind-numbing. We had 210 million two-wheelers and 70 million four-wheelers registered as on August 3, 2022. FY23 alone saw 22 million new vehicles, including 16 million two-wheelers and 3.6 million passenger vehicles. Vehicular emission is now the number one cause of pollution in many of India's major cities. The exhaust pipes of vehicles emit a deadly cocktail of fumes and particulate matter that include dangerous oxides of nitrogen and sulphur besides air toxics such as carbon monoxide and benzene.

To address the issue, India has since 2000 been following the Bharat Stage (BS) emission standards, modelled on the EU norms. The BS standards regulate tailpipe emissions of air pollutants. In April 2020, India leapfrogged from BS-IV to BS-VI implementation, with the current standards similar to EU-6/VI norms. Under the new norms,



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